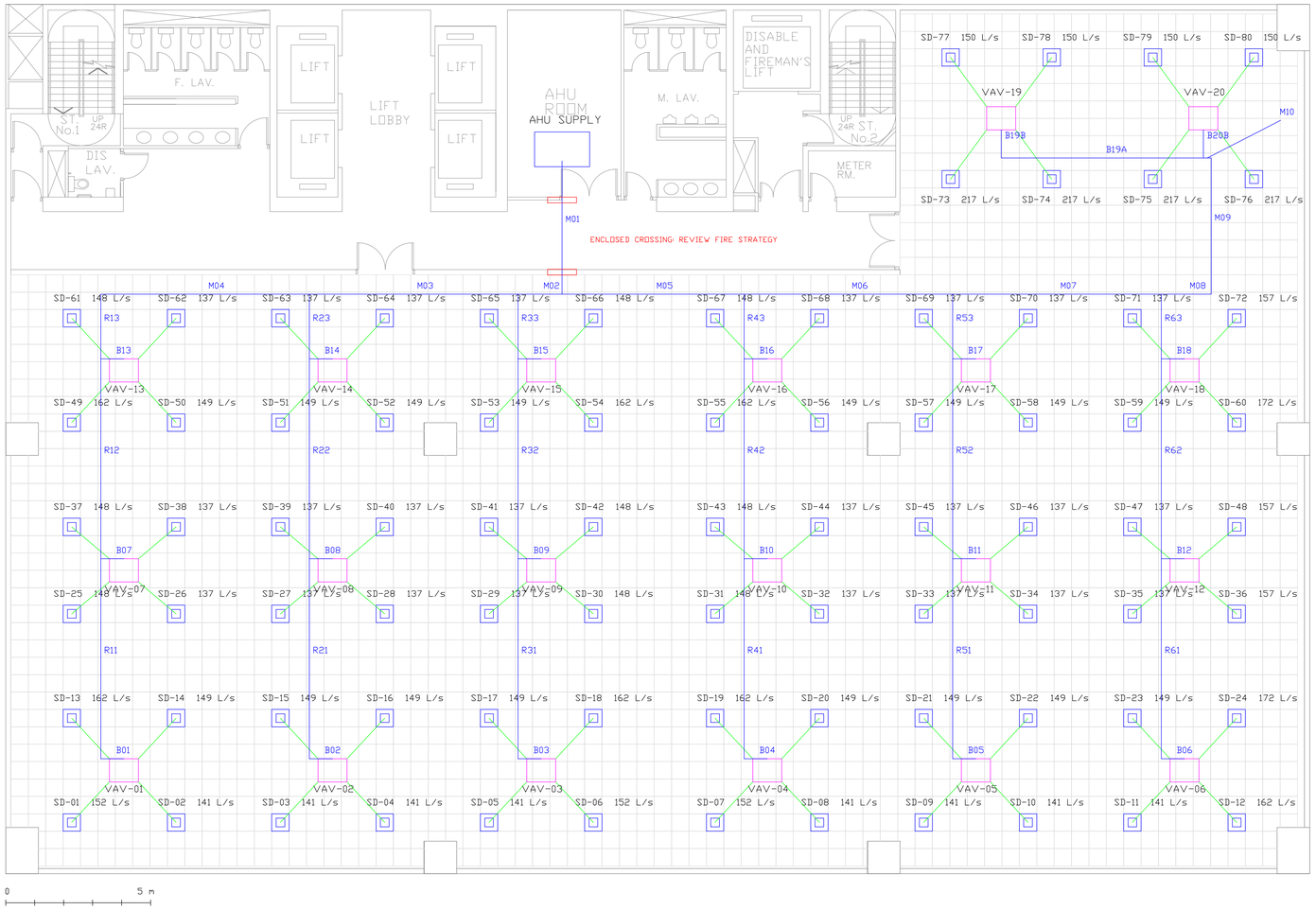


BSE3227 / HVACR 1

HVAC COURSE DESIGN v06

FOR USER ACCEPTANCE - COURSE DESIGN

80 SUPPLY DIFFUSERS / 20 VAV BOXES

4 outlets per VAV box

Printed office area: 1022 m2

Course total load: 189 W/m2

Course SHR: 0.90; room/supply 24/12 C

Peak supply: 12000.4 L/s

Minimum supply: 4800.1 L/s

BLUE: rigid supply duct / diffuser

GREEN: flexible connection

MAGENTA: VAV (schematic envelope)

All flex routes include 0.50 m allowance.

Maximum calculated flex: 2.607 m

Flex limit: 4 m (FSD 2023 cl. 7.2)

Assignment limit: 5 m

Duct sizes: width x height, mm

Main velocity <= 10.0 m/s

Branch velocity <= 8.0 m/s

Straight rigid friction <= 0.8 Pa/m

Rectangular aspect ratio <= 4:1

COURSE ADPI: T0.25/L = 1.0 - 3.4

Max / 40% flow: course checks pass.

Manufacturer classification differs.

See course-method qualification.

Ceiling / beam clearances assumed;

coordinate before construction.

AHU corridor crossing requires fire

enclosure / boundary protection review.

Source architecture retained.

Upper 8 outlets: coordinate grid.

RIGID SUPPLY DUCT SCHEDULE

ID	L/s	W x H (mm)	m/s	Pa/m
M01	12000.4	1350 x 900	9.88	0.754
M02	5238.2	900 x 700	8.31	0.794
M03	3468.5	850 x 550	7.42	0.781
M04	1769.7	650 x 450	6.05	0.701
M05	6762.2	900 x 850	8.84	0.786
M06	4992.5	950 x 650	8.09	0.771
M07	3293.7	900 x 500	7.32	0.798
R13	1769.7	650 x 450	6.05	0.701
B13	595.9	350 x 350	4.86	0.776
R12	1173.7	450 x 450	5.80	0.792
B07	570.0	400 x 300	4.75	0.761
R11	603.7	350 x 350	4.93	0.795
B01	603.7	350 x 350	4.93	0.795
R23	1698.9	600 x 450	6.29	0.785
B14	572.1	400 x 300	4.77	0.767
R22	1126.8	500 x 400	5.63	0.762
B08	547.2	400 x 300	4.56	0.706
R21	579.6	400 x 300	4.83	0.785
B02	579.6	400 x 300	4.83	0.785
R33	1769.7	650 x 450	6.05	0.701
B15	595.9	350 x 350	4.86	0.776
R32	1173.7	450 x 450	5.80	0.792
B09	570.0	400 x 300	4.75	0.761
R31	603.7	350 x 350	4.93	0.795
B03	603.7	350 x 350	4.93	0.795
R43	1769.7	650 x 450	6.05	0.701
B16	595.9	350 x 350	4.86	0.776
R42	1173.7	450 x 450	5.80	0.792
B10	570.0	400 x 300	4.75	0.761
R41	603.7	350 x 350	4.93	0.795
B04	603.7	350 x 350	4.93	0.795
R53	1698.9	600 x 450	6.29	0.785
B17	572.1	400 x 300	4.77	0.767
R52	1126.8	500 x 400	5.63	0.762
B11	547.2	400 x 300	4.56	0.706
R51	579.6	400 x 300	4.83	0.785
B05	579.6	400 x 300	4.83	0.785
R63	1826.3	650 x 450	6.24	0.743
B18	615.0	450 x 300	4.56	0.664
R62	1211.3	550 x 400	5.51	0.695
B12	588.3	350 x 350	4.80	0.758
R61	623.0	450 x 300	4.61	0.680
B06	623.0	450 x 300	4.61	0.680
M08	1467.4	550 x 450	5.93	0.736
M09	1467.4	550 x 450	5.93	0.736
M10	1467.4	550 x 450	5.93	0.736
B19A	733.7	500 x 300	4.89	0.721
B19B	733.7	500 x 300	4.89	0.721
B20B	733.7	500 x 300	4.89	0.721

Sizes apply to section IDs shown on plan.

Pressure gradient excludes fittings / flex.

TYPICAL FLOOR / SUPPLY AIR SINGLE-LINE PLAN

HV-01 | A1 1:100 | 2026-11-09 | USER ACCEPTANCE v06